ICT TECHNICIAN L

ICT/OS/IT/CC/01/6

Apply Basic Electronics Skills

WRITTEN ASSESSMENT

TIME: 2 HOURS

INSTRUCTIONS TO CANDIDATE

i. *This paper consists of **TWO** sections: **A** and **B**.*
ii. *Answer **ALL** questions in section A and **ANY THREE** (3) questions in section B.*
iii. *Marks for each question are indicated in the brackets.*
iv. *Candidates are provided with a separate answer booklet*
v. *Do not write on the question paper.*

SECTION A (40 MARKS)

Answer ALL questions in this section

1. A technician is troubleshooting a simple series circuit supplying a light bulb and a resistor. Outline a tool that can be used to determine if the circuit is complete. (2 Marks

2. During a lab session, a student measures the current through a resistor in a circuit and finds it lower than expected. Identify FOUR possible causes related to circuit configuration. (4 Mar

3. A diode is connected in a circuit and does not allow current to flow in one direction. U semiconductor theory, identify the phenomenon causing this behavior. (2 M

4. A student is asked to identify components in a simple electric circuit. State FOUR electronic components commonly used in circuits. (

5. A computer engineer is designing a buffer for data storage. Classify TWO types o suitable for temporary and permanent storage.

12. a) A trainee is required to work with different number systems while developing software for a digital calculator. Explain FIVE disadvantages of using Binary Coded Decimal (BCD) in computers for arithmetic operations. (10 Marks)

b) The rapid development of electronics components has changed the face of the electronic world over the past few years. Explain FIVE emerging trends of such modern electronic components. (10 Marks)

13.

a) Explain THREE ways number systems are applied in computer systems. (6 Marks)

b) Perform the following number systems conversion

156_{10} to: (3 Marks)

i. Binary (2 Marks)

ii. Octal

c) Perform the following number systems arithmetics (2 Marks)

i. $11011_2 + 10110_2$ (4 Marks)

ii. $1101_2 - 1011_2$ using 2's Complement (3 Marks)

iii. $101_2 \times 11_2$

14. An ICT technician is servicing several computers with suspected power-related faults.

a) Explain how voltage, current, and resistance measurements are used together to diagnose electronic problems. (6 Marks

b) Describe FOUR risks of incorrect use of electronic measuring instruments in IC maintenance. (8 Mar

c) Propose THREE corrective actions to ensure accurate electronic measurements in futu tasks. (6 M

15.

a) Outline THREE differences between N-type and P-type semiconductors. (6

b) Explain the following terms as used in Bipolar transistor configuration. (6

i. Common Base Configuration

ii. Common Emitter Configuration

iii. Common Collector Configuration

c) Discuss FOUR types of Read Only Memory.

Heather

3.

1. Multimeter
2. Weak power supply
   Loose connections
   High resistance in the circuit
   Incorrect resistor value.
3. Formation of a depletion region.
   The potential barrier blocks current in reverse bias.
4. Resistor
   Capacitor
   Diode
   Transistor
5. RAM - Volatile memory
   ROM - Permanent memory
6. 45 = 00101101 Show the working
7. First NAND gate $X = (A \cdot B)'$
   Second NAND gate $Y = (X \cdot X)'$
8. $I = \frac{V}{R}$

   $I = \frac{9}{470}$ $I = 0.01914\,A = 19.1\,mA$

9. (a) Inspect the PCB for damaged tracks.
   Use a multimeter to test continuity.

10. 3 = 0011

F = 1111

= 00111111₂

11. R = 100 + 200 + 300 = 600Ω

$I = \frac{12}{600} = 0.02$ A

V = IR = 0.02 × 100

= 2V

Section B.

12(a) Uses more bits than binary

Slower arithmetic operations

Requires complex conversion

Needs more processing hardware.

Requires more memory space.

(b) Energy efficiency - Components consume less power.

IoT - Components can connect to the internet.

Miniaturization - Components are smaller.

Wearable electronics - Development of fitness trackers.

13(a) Binary - Used in computer processing.

Octal - Used as shorthand for binary numbers.

Hexadecimal - Used in programming.

(b)(i) 156 = 10011100₂

(ii) 156 = 234₈

&

(c) (i) 11011₂ + 10 110₂

= 110001₂

Show the working!!

(ii) $1101_2 - 1011_2$ using 2's complement.

2's complement of $1011 = 0101$

$1101 + 0101 = 10010$

$= 0010_2$

arrange Your work!

(iii) $101_2 \times 11_2$

$= 1111_2$

15(a) N-type

Uses pentavalent impurities

Electrons are majority carriers

Donor impurities are used.

P-type

Uses trivalent impurities.

Holes are majority carri

Acceptor impurities are

(b) (i) Common base - Base terminal is common to input and o circuits.

(ii) Common emitter - Emitter terminal is common to input and output circuits.

(iii) Common collector - Collector terminal is common to inpu and output circuits.

(c) Mask ROM - programmed during manufacturing.

PROM - Can be programmed once.

EPROM - Can be erased using ultraviolet light and re

EEPROM - Can be erased and reprogrammed electrically.